

Question # 1:-

#1

$$\sum_{i=1,3,9}^{n-1} \sum_{j=0}^i 1 = \sum_{i=1,3,9}^{n-1} (i+1)$$

$$i = 1, 3, 9, \dots, n \Rightarrow i = 3^k, k = 0, 1, 2, \dots, \log_3 n$$

$$\Rightarrow \sum_{k=0}^{\log_3 n} (3^k + 1) \Rightarrow \sum_{k=0}^{\log_3 n} 3^k + \sum_{k=0}^{\log_3 n} 1$$

$$\Rightarrow \text{Geometric Series} \quad \because r > 1 \quad \because a = 1$$

$$\because 3^0 = 1$$

$$\Rightarrow \frac{a(r^n - 1)}{r - 1} \quad \because \text{no. of terms} = \log_3 n + 1$$

$$\Rightarrow \frac{3^{\log_3 n + 1} - 1}{3 - 1} \Rightarrow \frac{3^{\log_3 n} \cdot 3^1 - 1}{2}$$

$$\Rightarrow \frac{n \cdot 3 - 1}{2} \Rightarrow \frac{3n - 1}{2}$$

$$T(n) \approx \frac{3n - 1}{2} \Rightarrow \boxed{T(n) \approx n}$$

#2

$$\Rightarrow \sum_{i=1,2,4}^n \sum_{j=1,2,4}^i 1$$

$$\Rightarrow \substack{j=1,2,4,\dots,i \\ \log_2 i} \Rightarrow j=2^k \therefore k=0,1,\dots,\log_2 i$$

$$\Rightarrow \sum_{k=0,1,2}^{\log_2 i} 1 \Rightarrow (\log_2 i + 1)$$

$$\Rightarrow \sum_{i=1,2,4}^n (\log_2 i + 1)$$

$$\Rightarrow i=1,2,4,\dots,n \Rightarrow i=2^k ; k=0,1,\dots,\log_2 n$$

$$\Rightarrow \sum_{k=0}^{\log_2 n} (\log_2 2^k + 1) \Rightarrow \sum_{k=0}^{\log_2 n} (k+1)$$

$$\Rightarrow \sum_{k=0}^{\log_2 n} k + \sum_{k=0}^{\log_2 n} 1 \Rightarrow \frac{(\log_2 n)(\log_2 n + 1)}{2}$$

$$+ (\log_2 n + 1)$$

$$\Rightarrow \frac{(\log_2 n)^2 + (\log_2 n)}{2} + \log_2 n + 1$$

$$\Rightarrow \approx \frac{2}{2} (\log n)^2 + \log n + \log n + 1$$

$$\Rightarrow \boxed{T(n) \approx O((\log n)^2)}$$

#3

$$\Rightarrow \sum_{i=1,2,4}^n \sum_{j=i,2i,3i}^n \sum_{k=j,j/2,j/4}^0 1$$

$$\sum_{k=j,j/2,j/4}^0 1 \Rightarrow \sum_{k=1,2,4}^j 1$$

$$k = 1, 2, 4, \dots, j \Rightarrow k = 2^p; p = 0, 1, 2, \dots, \log_2 j$$

$$\Rightarrow \sum_{p=0}^{\log_2 j} 1 \Rightarrow \log_2 j + 1$$

$$\Rightarrow \sum_{j=i,2i,3i}^n (\log_2 j + 1)$$

$$j = i, 2i, 3i, \dots, n; \text{ let } m = 1, 2, 3, \dots, n/i$$

$$j = mi \Rightarrow n \quad j \rightarrow n \quad \therefore m \cdot i = n$$

$$m = n/i$$

$$\Rightarrow \sum_{m=1}^{n/i} (\log_2 (m \cdot i) + 1)$$

$$\Rightarrow \sum_{m=1}^{n/i} (\log_2 m + \log_2 i + 1)$$

$$\Rightarrow \sum_{m=1}^{n/i} \log_2 m + \log_2 i \sum_{m=1}^{n/i} 1 + \sum_{m=1}^{n/i} 1$$

$$\Rightarrow \log 1 + \log 2 + \dots + \log_{2i} \frac{n}{2i} \approx \log \left(\frac{n!}{i!} \right)$$

$$\Rightarrow \frac{n}{i} \log_{2i} \left(\frac{n}{2i} \right) + \log_{2i} i \left(\frac{n}{i} - \cancel{1} + \cancel{1} \right) + \left(\frac{n}{i} - \cancel{1} + \cancel{1} \right)$$

$$\Rightarrow \frac{n}{i} \log_2 \left(\frac{n}{i} \right) + \log_2 i \left(\frac{n}{i} \right) + \frac{n}{i}$$

$$\Rightarrow \frac{n}{i} \left(\log_2 \left(\frac{n}{i} \right) + \log_2 i \right) + \frac{n}{i}$$

$$\Rightarrow \sum_{i=1,2,4}^n \left(\frac{n}{i} \left(\log_2 \left(\frac{n}{i} \right) + \log_2 i \right) + \frac{n}{i} \right)$$

$$\Rightarrow n \sum_{i=1,2,4}^n \frac{1}{i} \left(\left(\log_2 n - \log_2 i \right) + \log_2 i \right) + n \sum_{i=1,2,4}^n \frac{1}{i}$$

$$\Rightarrow n \sum_{i=1,2,4}^n \frac{\log_2 n}{i} - \frac{\log_2 i}{i} + \frac{\log_2 i}{i} + n \sum_{i=1,2,4}^n \frac{1}{i}$$

$$\Rightarrow n \log_2 n \sum_{i=1,2,4}^n \frac{1}{i} + n \sum_{i=1,2,4}^n \frac{1}{i}$$

$$i = 1, 2, 4, \dots, n \Rightarrow i = 2^k; k = 0, 1, 2, \dots, \log_2 n$$

$$\Rightarrow n \log_2 n \sum_{k=0}^{\log_2 n} \frac{1}{2^k} + n \sum_{k=0}^{\log_2 n} \frac{1}{2^k}$$

Geometric Series: - $\sum_{k=0}^{\log_2 n} \frac{1}{2^k} \quad \therefore r = \frac{1}{2}$

$r < 1 \quad \therefore r = 1/2$

$\Rightarrow \frac{a(1-r^n)}{1-r} \quad a = \frac{1}{2^0} = \frac{1}{2} = 1$

$n = \log_2 n + 1$

$\Rightarrow \frac{1 \left(1 - \left(\frac{1}{2} \right)^{\log_2 n + 1} \right)}{1 - \frac{1}{2}}$

$\Rightarrow \frac{1 - \frac{1}{2^{\log_2 n + 1}}}{\frac{1}{2}} \Rightarrow 2 \left(1 - \frac{1}{2^{\log_2 n + 1}} \right)$

$\Rightarrow 2 \left(1 - \frac{1}{2^{\log_2 n} \cdot 2} \right) \Rightarrow 2 \left(1 - \frac{1}{2n} \right)$

$\Rightarrow 2 - \frac{1}{n}$ for asymptotic $n \approx \infty$

$\Rightarrow 2 - \frac{1}{\infty} \Rightarrow 2 - 0 \Rightarrow 2$

$\Rightarrow O(1)$

$\Rightarrow n \log_2 n \left(O(1) + nO(1) \right) \Rightarrow n \log_2 n$

$$\Rightarrow \boxed{T(n) \approx O(n \log n)}$$

#4

$$\Rightarrow \sum_{i=1,2,4}^n \sum_{j=1,2,4}^i 1$$

$$\therefore \sum_{j=1,2,4}^i 1$$

$$\therefore j=1,2,4,\dots,i \Rightarrow j=2^k$$

$$\therefore k=0,1,2,\dots,\log_2 i$$

$$\Rightarrow \sum_{k=0}^{\log_2 i} 1 \Rightarrow \log_2 i + 1$$

$$\Rightarrow \sum_{i=1,2,4}^n (\log_2 i + 1)$$

$$\Rightarrow \sum_{i=1,2,4}^n \log_2 i + \sum_{i=1,2,4}^n 1$$

$$i=1,2,4,\dots,n \Rightarrow i=2^k ; k=0,1,2,\dots,\log_2 n$$

$$\Rightarrow \sum_{k=0}^{\log_2 n} \log_2 2^k + \sum_{k=0}^{\log_2 n} 1$$

$$\Rightarrow \sum_{k=0}^{\log_2 n} k + (\log_2 n + 1)$$

$\Rightarrow$

$$\Rightarrow \frac{(\log_2 n)(\log_2 n + 1)}{2} + \log_2 n + 1$$

$$\Rightarrow \frac{(\log_2 n)^2 + \log_2 n}{2} + \log_2 n + 1$$

$$\Rightarrow (\log_2 n)^2 + \log_2 n + 2\log_2 n + 2$$

$$\approx (\log_2 n)^2 \approx (\log n)^2$$

$$\Rightarrow \boxed{T(N) = (\log n)^2}$$

#5

$$\Rightarrow \sum_{i=0}^{n-1} \sum_{j=0}^{n-i-1} 1$$

$$\Rightarrow \sum_{i=0}^{n-1} (n-i-0+1)$$

$$\Rightarrow \sum_{i=0}^{n-1} (n-i)$$

$$\Rightarrow \sum_{i=0}^{n-1} n - \sum_{i=0}^{n-1} i$$

$$\Rightarrow n \sum_{i=0}^{n-1} 1 - \sum_{i=0}^{n-1} i$$

$$\Rightarrow n(n - 1 - 0 + 1) - \left(\frac{(n-1)(n-1+1)}{2} \right)$$

$$\Rightarrow n(n) - \frac{n(n-1)}{2}$$

$$\approx n^2 \Rightarrow \boxed{T(n) \approx O(n^2)}$$

7

$$\Rightarrow \sum_{i=1,2,4}^n \sum_{x=1,2,4}^n \sum_{j=1,2,4}^x 1$$

$$\Rightarrow j=1,2,4, \dots, x \Rightarrow j=2^k; k=0,1, \dots, \log_2 x$$

$$\Rightarrow \sum_{k=0}^{\log_2 x} 1 \Rightarrow \log_2 x + 1$$

$$\Rightarrow \sum_{x=1,2,4}^n \log_2 x + 1$$

$$x=1,2,4, \dots, n \Rightarrow x=2^k; k=0,1, \dots, \log_2 n$$

$$\Rightarrow \sum_{k=0}^{\log_2 n} (\log_2 2^k + 1) \Rightarrow \sum_{k=0}^{\log_2 n} (k+1)$$

$$\Rightarrow \sum_{l=0}^{\log_2 n} l + \sum_{l=0}^{\log_2 n} 1$$

$$\Rightarrow \frac{(\log_2 n)(\log_2 n + 1)}{2} + \log_2 n + 1$$

$$\Rightarrow \frac{(\log_2 n)^2 + \log_2 n + 2\log_2 n + 2}{2}$$

$$\Rightarrow \approx (\log_2 n)^2 + 3\log_2 n + 2$$

$$\Rightarrow \approx (\log_2 n)^2$$

$$\Rightarrow \sum_{i=1,2,4}^n (\log_2 n)^2 \Rightarrow (\log_2 n)^2 \sum_{i=1,2,4}^n 1$$

$$\Rightarrow i=1,2,4,\dots,n \Rightarrow i=2^k; k=0,1,\dots,\log_2 n$$

$$\Rightarrow (\log_2 n)^2 \sum_{k=0}^{\log_2 n} 1 \Rightarrow (\log_2 n)^2 (\log_2 n + 1)$$

$$\Rightarrow (\log_2 n)^3 + (\log_2 n)^2 \approx (\log n)^3 + (\log n)^2$$

$$\Rightarrow \boxed{T(N) \approx O((\log n)^3)}$$

#8

$$d^2 \leq n \Rightarrow d \leq \sqrt{n}$$

Linear loop $\Rightarrow T(N) \approx O(\sqrt{n})$
for function sqrt

But in other function, each calls sqrt(n)
so each call gives $O(\sqrt{n})$ for single
execution of loop. Hence one (1) call
iterates $O(\sqrt{n})$ times and since $\leq \sqrt{n}$
it also moves $O(\sqrt{n})$ times. Nested
scenario takes place and hence:-

$$T(N) \approx O(\sqrt{n}) \times O(\sqrt{n}) \approx O(n)$$

Question no. 2:-

#1

$$\Rightarrow \sum_{i=0}^n \sum_{j=0}^i \sum_{k=0}^j 1$$

$$\Rightarrow \sum_{j=0}^i (j+1) = \sum_{j=0}^i j + \sum_{j=0}^i 1$$

$$\Rightarrow \frac{i(i+1)}{2} + (i+1)$$

$$\Rightarrow \frac{1}{2} \sum_{i=0}^n (i^2 + 3i + 2) \approx \sum_{i=0}^n (i^2 + 3i + 2)$$

$$\Rightarrow \approx \sum_{i=0}^n i^2 \Rightarrow \frac{n(n+1)(2n+1)}{2}$$

$$\Rightarrow \boxed{T(n) \approx O(n^3)} ; \boxed{\text{Space} \approx O(n^3)}$$

#2
simple loop $\rightarrow O(n)$
size * 2 $\rightarrow 2^n$

$\Rightarrow$ allocating ~~time~~ ^{Space} outside loop $\approx 2^n$

$$\Rightarrow O(n) + O(2^n) \approx O(2^n)$$

$$\boxed{T(N) \approx O(\cancel{2^n}) O(n)} ; \boxed{\text{Space} \approx O(2^n)}$$

#3
 $\Rightarrow$ loop outside $\rightarrow O(n)$

$$\Rightarrow \text{nested loop} \rightarrow \sum_{i=0}^n \sum_{j=0}^i 1 = \sum_{i=0}^n (i+1) \approx n^2$$

Allocation space time taken $\approx n^2$

$$\boxed{T(N) \approx O(n^2)} ; \boxed{\text{Space Complexity} \approx O(n^2)}$$

#6

Assuming $j \leq n$ is a typo and real statement is $i \leq n$, then we get:-

$$\Rightarrow \sum_{i=0}^n \sum_{j=0}^{i^2} \sum_{k=0}^j 1$$

$$\Rightarrow \sum_{k=0}^j 1 \Rightarrow (j+1)$$

$$\Rightarrow \sum_{j=0}^{i^2} (j+1) \Rightarrow \sum_{j=0}^{i^2} j + \sum_{j=0}^{i^2} 1$$

$$\Rightarrow \frac{i^2(i^2+1)}{2} + i^2 + 1$$

$$\Rightarrow \frac{i^4 + i^2 + i^2 + 1}{2} \Rightarrow \approx i^4$$

$$\Rightarrow \sum_{i=0}^n i^4 \Rightarrow \frac{n(n+1)(2n+1)(3n^2+3n+1)}{30}$$

$$\Rightarrow \approx n^5 \Rightarrow \boxed{T(N) \approx O(n^5)}$$